



Store at: +2+8° C.

### Presentation:

Cod. SU037-SP CONT: R1 1 x 40mL. R2 1 x 10 mL. + CAL 1 x 5 mL

Cod. SU036 CONT: R1 1 x 100 mL. R2 1 x 25 mL. + CAL 1 x 5 mL

Cod. SU039 CONT: R1 2 x 100 mL. R2 2 x 25 mL. + CAL 1 x 5 mL

## Procedure

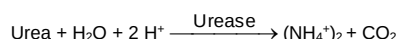
### Quantitative determination of urea

#### Only for in vitro use in clinical laboratory (IVD)

#### TEST SUMMARY

Urea in the sample is hydrolyzed enzymatically into ammonia (NH<sub>4</sub><sup>+</sup>) and carbon dioxide (CO<sub>2</sub>).

Ammonia ions formed reacts with α-ketoglutarate in a reaction catalysed by glutamate dehydrogenase (GLDH) with simultaneous oxidation of NADH to NAD<sup>+</sup>:



The decrease in concentration of NADH, is proportional to urea concentration in the sample<sup>1</sup>.

#### REAGENTS COMPOSITION

|                       |                                          |                                    |
|-----------------------|------------------------------------------|------------------------------------|
| <b>R 1</b><br>Buffer  | TRIS pH 7.8<br>α-Ketoglutarate<br>Urease | 80 mmol/L<br>6 mmol/L<br>75000 U/L |
| <b>R 2</b><br>Enzymes | GLDH<br>NADH                             | 60000 U/L<br>0.32 mmol/L           |
| <b>UREA CAL</b>       | Urea aqueous primary standard 50 mg/dL   |                                    |

#### REAGENT PREPARATION AND STABILITY

Working reagent (WR)

Mix: 4 vol. R1 Buffer + 1 vol. R2 Substrate.

The (WR) is stable for 1 month at 2-8°C or 1 week at room temperature (15-25°C).

**UREA CAL:** Ready to use.

Proceed carefully with this product because due its nature it can get contaminated easily.

Signs of reagent deterioration:

- Presence of particles and turbidity.
- Blank absorbance (A) at 340 nm < 1.00.

**All the reagents of the kit are stable up to the end of the indicated month and year of expiry. Store tightly closed at 2-8°C, protected from light and contaminations prevented during their use. Do not use reagents over the expiration date.**

#### SPECIMEN

- Serum or heparinized plasma<sup>1</sup>: Do not use ammonium salts or fluoride as anticoagulants.
- Urine<sup>1</sup>: Dilute sample 1/50 in distilled water. Mix. Multiply the results by 50 (dilution factor). Preserve urine samples at pH < 4. Urea is stable at 2-8°C for 5 days.

#### MATERIAL REQUIRED BUT NOT PROVIDED

- Spectrophotometer or colorimeter measuring at 340 nm.
- Matched cuvettes 1.0 cm light path.

#### General laboratory equipment (Note 1)

#### TEST PROCEDURE

- Assay conditions:
  - Wavelength: ..... 340 nm
  - Cuvette: ..... 1 cm light path
  - Temperature: ..... 37°C / 15-25°C
- Adjust the instrument to zero with distilled water.
- Pipette into a cuvette:

|                          | Blank | Standard | Sample |
|--------------------------|-------|----------|--------|
| WR (mL)                  | 1.0   | 1.0      | 1.0    |
| Standard (μL) (Note 2,3) | --    | 10       | --     |
| Sample (μL)              | --    | --       | 10     |

- Mix and read the absorbance after 30 s (A<sub>1</sub>) and 90 s (A<sub>2</sub>). Calculate: ΔA = A<sub>1</sub> - A<sub>2</sub>

#### CALCULATIONS

$$\frac{(A_1 - A_2) \text{ Sample} - (A_1 - A_2) \text{ Blanco}}{(A_1 - A_2) \text{ Calibrator} - (A_1 - A_2) \text{ Blanco}} \times 50 \text{ (Calibrator conc)} = \text{mg/dL urea}$$
  
in the sample

10 mg/L urea BUN divided by 0.466 = 21 mg/L urea = 0.36 mmol/L urea<sup>1</sup>.

**Conversion factor:** mg/dL x 0.1665 = mmol/L.

#### QUALITY CONTROL

Control sera are recommended to monitor the performance of the procedure, Control H Normal Ref. QC003 and Control H Pathological Ref. QC004. If control values are found outside the defined range, check the instrument, reagents and calibrator for problems.

**Serum controls are recommended for internal quality control. Each laboratory should establish its own Quality Control scheme and corrective actions**

#### REFERENCE VALUES <sup>5,6</sup>

Serum or plasma:

15-45 mg/dL ≅ 2.5-7.5 mmol/L

Urine:

26 - 43 g/24 h ≅ 428-714 mmol/24h

(These values are for orientation purpose).

**It is suggested that each laboratory establish its own reference range.**

#### CLINICAL SIGNIFICANCE

Urea is the final result of the metabolism of proteins; It is formed in the liver from their destruction.

It can appear the urea elevated in blood (uremia) in: diets with excess of proteins, renal diseases, heart failure, gastrointestinal hemorrhage, dehydration or renal obstruction<sup>1,4,5</sup>.

Clinical diagnosis should not be made on a single test result; it should integrate clinical and other laboratory data.

#### REAGENT PERFORMANCE

- **Measuring range:** From detection limit 0.743 mg/dL to linearity limit 400 mg/dL.

If the concentration is greater than linearity limit dilutes 1:2 the sample with ClNa 9 g/L and multiply the result by 2.

- **Precision:**

|              | Intra-assay (n=20) |      | Inter-assay (n=20) |      |
|--------------|--------------------|------|--------------------|------|
| Mean (mg/dL) | 37.5               | 120  | 40.0               | 126  |
| SD           | 1.05               | 0.92 | 1.06               | 2.07 |
| CV (%)       | 2.79               | 0.77 | 2.65               | 1.65 |

- **Sensitivity:** 1 mg/dL = 0.00180 A.

- **Accuracy:** Results obtained using GPL reagents (y) did not show systematic differences when compared with other commercial reagent (x).

The results obtained using 50 samples was the following:

Correlation coefficient (r)<sup>2</sup>: 0.98209.

Regression equation y = 1.0343x - 1.2105

The results of the performance characteristics depend on the analyzer used.

#### INTERFERING SUBSTANCES

It is recommended to use heparin as anticoagulant. Do not use ammonium salts or fluoride<sup>1</sup>.

A list of drugs and other interfering substances with urea determination has been reported <sup>2,3</sup>.

#### NOTES

- Glassware and distilled water must be free of ammonia and ammonium salts<sup>1</sup>.
- Calibration with the aqueous standard may cause a systematic error in automatic procedures. In these cases, it is recommended to use a serum Calibrator.
- Use clean disposable pipette tips for its dispensation.

#### BIBLIOGRAPHY

- Kaplan A. Urea. Kaplan A et al. Clin Chem The C.V. Mosby Co. St Louis. Toronto. Princeton 1984; 1257-1260 and 437 and 418.
- Young DS. Effects of drugs on Clinical Lab. Tests, 4th ed AACC Press, 1995.
- Young DS. Effects of disease on Clinical Lab. Tests, 4th ed AACC 2001.
- Burtis A et al. Tietz Textbook of Clinical Chemistry, 3rd ed AACC 1999.
- Tietz N W et al. Clinical Guide to Laboratory Tests, 3rd ed AACC 1995.



CHEMELEX, S.A.  
Pol. Ind. Can Castells. C / Industria 113, Nave J  
08420 Canovelles -BARCELONA-  
Tel- 34 93 849 17 35 Fax- 34 93 846 78 75



GPLBSDTT47-I  
Rev. 7 - 16/05/19